

CCDS-431 Social Web Analytics

Detecting Fake Influencers Using Multi-Layer Social Media Analysis

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Name	Role
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Introduction

Social media influencers are really important for marketing. A lot of businesses use social media influencers to tell people about their products and services. The thing is, not all social media influencers are real. Some social media influencers have people following them and they make it look like people are talking about their posts, when they are not. This makes it harder for businesses to get results from their marketing plans. So it is very important to look at the information about media influencers to make sure businesses are making good choices, about who they work with. Social media influencers can be a help but only if they are real social media influencers.

Business Problem

Lots of businesses pick influencers because they have a lot of followers. This is not always a good idea. Some influencers do not really have that followers they just use computers to make it look like they do. This means businesses might waste their money on marketing that does not work. The big issue, with influencers is figuring out which ones are influencers, and which ones are fake influencers. Businesses need to know which influencers really popular and which influencers are not.

Research Question How can businesses detect fake influencers using multi-layer social media analysis?

Dataset Description

The dataset has information about social media influencer accounts. How they behave. It has things like how followers they have how many people they follow the ratio of followers to people they follow how many posts they make, how often they post, how many spam comments they get how many regular comments they get if they have suspicious links, in their bio if they follow and unfollow people a lot what people think of them and if they are fake or not.

The dataset was made for school to show what influencers do on media like what they say what they do, who they talk to and what links they share. It was used to look at influencers who might be doing something and to help find out who is a fake influencer.

Methodology

I used a dataset that I found on Kaggle. This dataset has 1,773 rows and 27 features that are, about what influencers do. The original dataset had information from lots of media platforms. For this project I only used the features that are related to Instagram and X. I also added some columns to the dataset like text, sentiment score and sentiment label to help with the text analysis.

I cleaned up the dataset by looking for missing values and duplicate records. After preprocessing, no missing or duplicate values remained. I used Python and Google Colab to get the dataset ready and to analyze the influencer behavior dataset.

The project focused on four main layers:

Text Layer: Analyzed spam comment rate, generic comments, and sentiment analysis results to examine comment authenticity and repetitive behavior.

Action Layer: Examined engagement-related behavior such as followers, following, posting activity, and interaction patterns.

Network Layer: Focused on follower-following patterns and follow-unfollow activity to identify unusual account behavior.

Hyperlink Layer: Analyzed suspicious or promotional links included in influencer profiles.

Data Analysis

We looked at real and fake influencer behavior on media in a lot of different ways.

The analysis looked at real and fake influencer behavior. It focused on things, like spam comments and the patterns of who is following who.

It also looked at how real and fake influencers post things how people feel about what they post and if they share any suspicious links.

Real and fake influencer behavior was also compared by looking at who they follow and then unfollow.

Text Layer Analysis

The Text Layer analysis focused on sentiment labels, spam comment rates, and text patterns. The sentiment distribution graph compared real and fake accounts across positive, neutral, and negative sentiment categories.

The results showed that positive sentiment appeared most frequently in both real and fake accounts. However, real accounts appeared more frequently across all sentiment categories. This suggests that sentiment analysis alone is not enough to detect fake influencers without combining other social media layers.

A word cloud was also generated for fake accounts. Common words included terms such as strong followers, balanced activity, genuine account, normal user, fake account, and suspicious behavior. The repetition of these terms suggests that fake accounts may use repetitive text patterns.

Network Layer Analysis

The Network Layer focuses on the number of people someone follows and the number of people who follow them on Instagram. When we compare this information, we see that real accounts and fake accounts have much the same pattern.

The findings revealed that the follower/following ratio alone is insufficient to differentiate between genuine and fake influencer accounts. Therefore, the Network Layer should be combined with other social media layers for more reliable detection.

Action Layer Analysis

The Action Layer looked at how people posted things. It used the number of posts per day to do this. What they found out was that real people posted a little often than fake people. Real people posted about 0.36 times per day on average. Fake people posted about 0.29 times per day on average.

This means that real influencers probably post things regularly than fake influencers. The Action Layer and its findings, on influencers and fake influencers are really important here. Real influencers seem to keep posting things consistently than fake influencers do.

Hyperlink Layer Analysis

The Hyperlink Layer looked at links that people put in their account biographies. When we looked at the results we saw that most accounts did not have links. Then we found out that some accounts whether they were real or fake did have these suspicious links. This means that suspicious links can be a clue to find influencers but we should also look at other things on social media to be sure, about the Hyperlink Layer and fake influencers and the Hyperlink Layer. The Hyperlink Layer is important to consider when looking at influencers and the Hyperlink Layer can help us.

Integrated Multi-Layer Analysis

A correlation matrix was made to see how all the chosen features relate to accounts. The analysis found that individual features don't strongly connect to account status.

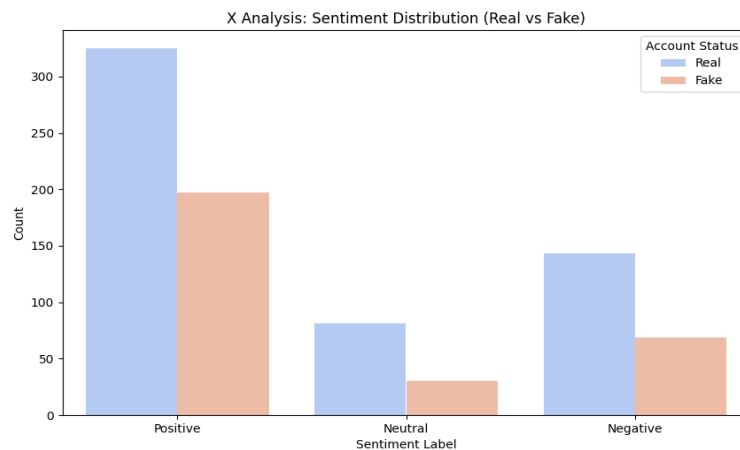
For example, the rate of spam comments was a bit positively related to fake accounts. On the hand the number of posts per day had a very weak negative connection to fake accounts. Other things like the ratio of followers to following links in bios and sentiment scores didn't show much connection at all. This means that using one feature can't accurately spot fake influencers. Combining social media aspects gives more reliable results, for detecting fake accounts. The fake account status and fake influencers are issues that need features to be addressed.

Results

- Text layer findings showed that fake accounts often contained repetitive text patterns and slightly higher spam comment activity.
- Action layer results suggested that real accounts generally maintained more consistent posting activity than fake accounts.
- The Network Layer results indicated that follower/following ratio is not a strong indicator by itself because both real and fake accounts showed similar distributions.
- Suspicious links appeared in both real and fake accounts, which suggests that this feature alone is not enough for accurate classification.
- The integrated analysis showed that individual features had weak relationships with fake account status. This suggests that combining multiple social media layers provides more reliable fake influencer detection than relying on a single metric.

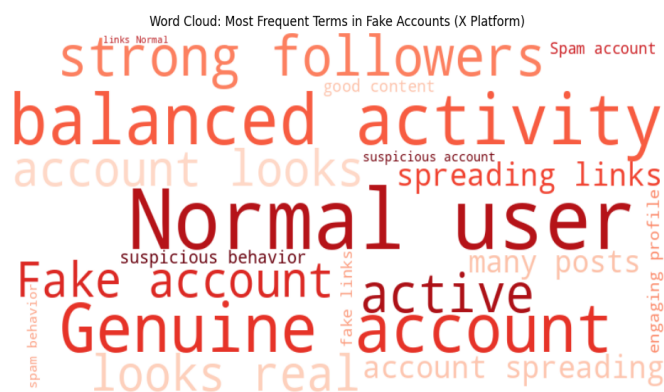
Visualizations

Figure 1: Sentiment Distribution of Real vs Fake Accounts



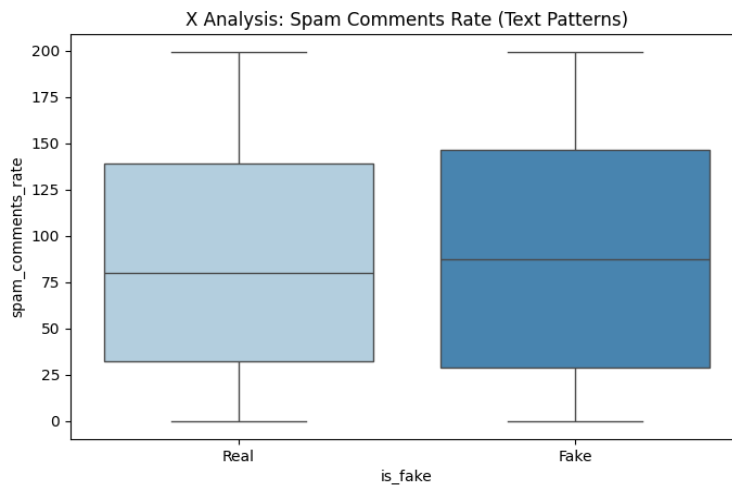
This figure compares positive, neutral, and negative sentiment between real and fake accounts on X. Positive sentiment appeared more frequently in both real and fake accounts. While real accounts showed higher overall counts.

Figure 2: Word Cloud for Fake Accounts



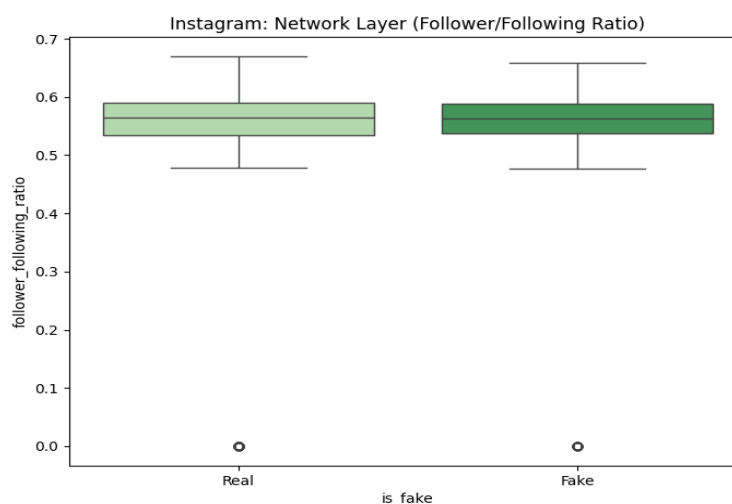
Common words related to followers, activity, and suspicious terms appeared frequently in fake account text patterns.

Figure 3: Spam Comments Rate Comparison



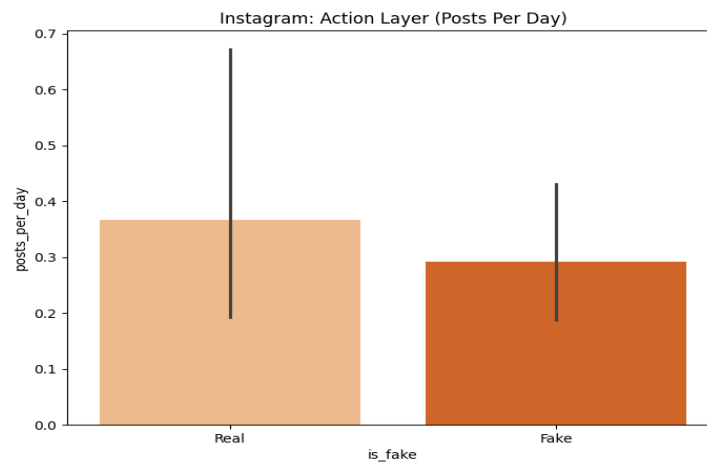
This boxplot compares spam comments rate between real and fake accounts. Fake accounts showed a slightly higher spam comments rate, suggesting more suspicious engagement behavior.

Figure 4: Follower/Following Ratio Comparison



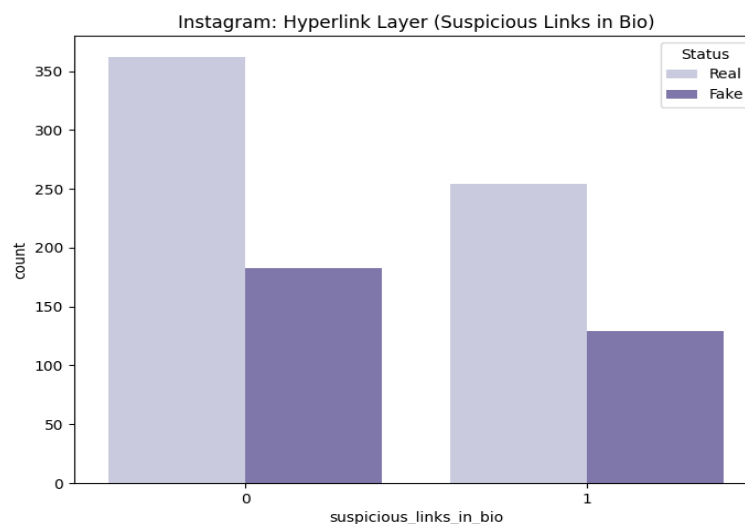
Real and fake accounts had similar follower and following ratios. This means that this parameter alone is insufficient for classifying accounts into real and fake accurately

Figure 5: Average Posts Per Day



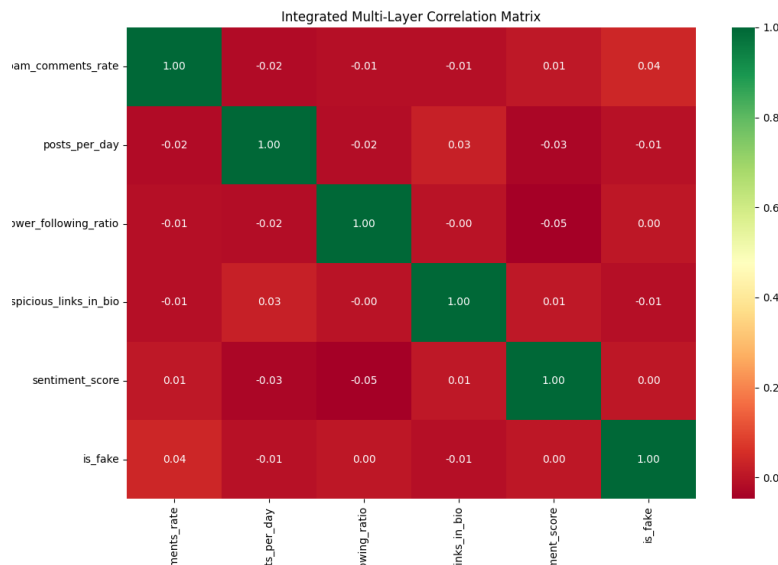
Real accounts showed slightly higher posting activity on average compared to fake accounts.

Figure 6: Suspicious Links in Bio



This figure shows the presence of suspicious links in account biographies. Although most accounts did not contain suspicious links, some real and fake accounts included them.

Figure 7: Correlation Matrix



The correlation matrix highlights weak relationships between individual features and fake account status.

Recommendations

Businesses should avoid relying only on follower count when selecting influencers for marketing campaigns.

Combining multiple social media layers provides more reliable fake influencer detection by analyzing indicators such as:

- Spam comment behavior
- Posting activity
- Sentiment patterns
- Suspicious links
- Network interaction behavior

Using a multi-layer analysis approach can help businesses select more authentic influencers, reduce marketing risks, and improve campaign effectiveness.

Conclusion

This project showed us how looking at media from many different angles can help us find fake influencers. The study looked at what people say what they do, who they talk to and what links they share to understand what influencers are like. We found out that fake accounts usually do a bit suspicious behaviour and they do not post things at the same time every day but you cannot just look at one thing to figure out if someone is fake or not. When we looked at how all these things are connected we saw that using many different ways to study influencers is better than just using one way to catch the fake ones. The project makes it clear that using data to make decisions, about influencer marketing is very important. It can help stop fake influencers from causing problems.

Appendix:

Original data set file: fake_social_media_global_2.0_with_missing.xlsx

Final processed data set: final_social_media_data.xlsx

The Python implementation code is included in the submitted zip file.

References:

Kaggle dataset: fake_social_media_global_2.0_with_missing.xlsx

Python libraries used in the project:

- Pandas
- Matplotlib
- Seaborn
- WordCloud